

Many other loose ends have been left in this brief introduction to modifiers. The astute reader may have noticed, for instance, that we have said nothing about PPs modifying verbs rather than nouns. A few cautiously chosen examples of this type happen to mean what we predict them to mean: for example, “Julius is sleeping on the couch” does seem to be true just in case Julius is sleeping and is on the couch. (*Exercise:* Show that this is what we predict.) Many other examples that readily come to mind, however, do not have the predicted meanings at all. Consider “John wrote on the blackboard”, “Mary put the book on the table”, or “Max tossed the salad in the bowl”, to name just a few. There are plenty of good grounds here to suspect that our present account of modifiers is very far from how modification really works in natural language. On the other hand, it might turn out that the problems lie elsewhere – for instance, in our simplistic assumptions about verb meaning and VP structure. Indeed, current research overwhelmingly points to the latter conclusion, and many *prima facie* counterexamples emerge as cases of intersective modification after all.¹⁰

4.4 The definite article

We have proposed that common nouns like “cat” denote the characteristic functions of sets of individuals. What does this imply for the semantic analysis of determiners? We will defer the general version of this question to chapter 6. Right here, we will look at only one determiner: namely, the definite article.¹¹

4.4.1 A lexical entry inspired by Frege

The basic intuition about phrases of the form “the NP” is that they denote individuals, just like proper names. Had it not been for Bertrand Russell’s famous claim to the contrary, few people would think otherwise. Frege, for one, thought it obvious: “let us start, e.g., with the expression ‘the capital of the German Empire.’ This obviously takes the place of a proper name, and has as its reference an object.”¹² Hence his practice of referring to definite descriptions as “compound proper names”.

If you read on in that particular passage, Frege imposes a rather odd syntactic analysis on his example. Instead of dividing it into the constituents “the” and “capital of the German Empire”, he splits it up into “the capital of” and “the German Empire”. He then proceeds to analyze “the capital of” as denoting a function from objects to objects (our type $\langle e, e \rangle$). Elsewhere, however, he treats a similar example as follows:

“the negative square root of 4”. We have here a case in which out of a concept-expression a compound proper name is formed with the help of the definite article in the singular, which is at any rate permissible when one and only one object falls under the concept.¹³

We follow Frege in this second syntactic and semantic analysis. Notice that by “concept-expression”, he means an expression whose meaning is of type $\langle e, t \rangle$. In his example, that’s the NP “negative square root of 4”, which indeed receives a meaning of that type if we analyze it along the lines of the previous sections of this chapter. (“Square root”, it seems, is a transitive noun like “part”, with a meaning of type $\langle e, \langle e, t \rangle \rangle$. “Of” is vacuous, $\llbracket \text{square root} \rrbracket$ applies to 4 by Functional Application, and the result of that composes with $\llbracket \text{negative} \rrbracket$ under Predicate Modification.)

The determiner “the”, then, denotes a function with arguments in $D_{\langle e, t \rangle}$ and values in D_e . For instance, $\llbracket \text{the} \rrbracket$ applied to the function $\llbracket \text{negative square root of 4} \rrbracket$ yields the number -2 . $\llbracket \text{the} \rrbracket$ applied to $\llbracket \text{president of the USA} \rrbracket$ yields Clinton at the time of writing. $\llbracket \text{the} \rrbracket$ applied to $\llbracket \text{opera by Beethoven} \rrbracket$ yields *Fidelio*. The generalization that emerges is (1).

- (1) For any $f \in D_{\langle e, t \rangle}$ such that there is exactly one x for which $f(x) = 1$, $\llbracket \text{the} \rrbracket(f) = \text{the unique } x \text{ for which } f(x) = 1$.

What about functions f which do *not* map exactly one individual to 1? What is $\llbracket \text{the} \rrbracket(f)$ for one of those?

Let’s examine our intuitions on this matter. What are the objects denoted by the following definites?

(2) **the escalator in South College**

(3) **the stairway in South College**

You should know that South College has no escalator and more than one stairway. Once we are aware of this, we are hard pressed to say which objects (2) and (3) denote. The only natural answer is that neither of these phrases denotes any object at all. Let’s go ahead and implement precisely this simple-minded intuition in our lexical entry for “the”.

What we are aiming to predict is that (2) and (3) have no semantic value. In other words, there is no such thing as $\llbracket \text{the escalator in South College} \rrbracket$ or $\llbracket \text{the stairway in South College} \rrbracket$. The reason has to be that the functions $\llbracket \text{escalator in South College} \rrbracket$ and $\llbracket \text{stairway in South College} \rrbracket$ are *not in the domain of* $\llbracket \text{the} \rrbracket$. If they are not in the domain of $\llbracket \text{the} \rrbracket$, then $\llbracket \text{the} \rrbracket$ can’t apply to them, and this means that we cannot apply FA to calculate a semantic value for the DP-nodes in (2) or (3). The generalization that emerges regarding the domain of $\llbracket \text{the} \rrbracket$ is this:

- (4) The domain of $\llbracket \text{the} \rrbracket$ contains just those functions $f \in D_{\langle e, t \rangle}$ which satisfy the condition that there is exactly one x for which $f(x) = 1$.

Putting (1) and (4) together, we can now formulate our lexical entry for “the”:

- (5) $\llbracket \text{the} \rrbracket =$
 $\lambda f : f \in D_{\langle e, t \rangle}$ and there is exactly one x such that $f(x) = 1$.
 the unique y such that $f(y) = 1$.

This is a bit of an unwieldy λ -term,¹⁴ but if you apply it to the examples above, you can see that it describes the function we were trying to define.

Before we end this subsection, let's dispose of a technical matter. What is the semantic type of $\llbracket \text{the} \rrbracket$? In the strict sense of our definitions so far, it actually has none. To say that its type is $\langle \langle e, t \rangle, e \rangle$ would mean that it is a function from $D_{\langle e, t \rangle}$ to D_e . But “from $D_{\langle e, t \rangle}$ ” means “with domain $D_{\langle e, t \rangle}$ ”, and we have just seen that the domain of $\llbracket \text{the} \rrbracket$ is not $D_{\langle e, t \rangle}$ but only a subset thereof. At this point, we find ourselves with an inconvenient terminology, and we will simply change it. We henceforth define $D_{\langle \sigma, \tau \rangle}$ (for any types σ, τ) as the set of all *partial* functions from D_σ to D_τ . “Partial function from” is defined as follows:

- (6) A *partial function from* A to B is a function from a subset of A to B .

(When we emphatically mean “function from” rather than “partial function from”, we will sometimes say “*total* function from”.) With these new definitions, we can now say that $\llbracket \text{the} \rrbracket$ is in $D_{\langle \langle e, t \rangle, e \rangle}$, or that its type is $\langle \langle e, t \rangle, e \rangle$.

4.4.2 *Partial denotations and the distinction between presupposition and assertion*

When a tree contains a lexical item that denotes a partial function, this may cause the tree to wind up without a semantic value. We have already seen this happen in (2) and (3) above. In larger structures, there are repercussions all the way up the tree. For instance, the following sentences are predicted not to have any semantic values, neither 1 nor 0 nor anything else.

- (7) The stairway in South College is dirty.
 (8) John is on the escalator in South College.

This is a direct consequence of our set of composition principles. The only principle that could potentially provide a semantic value for the branching node

above the definite description (that is, the S-node in (7) and the PP-node in (8)) is Functional Application, repeated here from chapter 3.

(9) *Functional Application* (FA)

If α is a branching node and $\{\beta, \gamma\}$ is the set of α 's daughters, then α is in the domain of $\llbracket \cdot \rrbracket$ if both β and γ are and $\llbracket \gamma \rrbracket$ is in the domain of $\llbracket \beta \rrbracket$. In this case, $\llbracket \alpha \rrbracket = \llbracket \beta \rrbracket(\llbracket \gamma \rrbracket)$.

But, as this formulation makes plain, you can't apply the denotation of one daughter to that of the other unless both daughters *have* denotations. So FA can't apply (and no other composition principle even came close). By the same reasoning, no semantic values can be obtained for any higher nodes that indirectly dominate a denotationless definite description like (2) or (3).

Exercise

Consider an example with one definite description embedded in another:

(i) The killer of the black cat escaped.

(a) Draw an interpretable syntactic structure for (i).

(b) Describe three possible states of affairs:

one where (i) is false;

another one where (i) lacks a truth-value because "the black cat" has no extension;

a third one where (i) is also without a truth-value, but this time because "the killer of the black cat" lacks a denotation. (Assume for this third scenario that "the black cat" does have a denotation.)

Are the empirical predictions that are implied by our current semantic component correct? Consider what we predict about (8): If you know English, and if you furthermore know that there is no escalator in South College, then you know that:

(a) the sentence "John is on the escalator in South College" is not true,
and

(b) the sentence "John is on the escalator in South College" is not false.

Part (a) of this prediction is unobjectionable. But in apparent disagreement with (b), many informants will spontaneously classify the assertion "John is on the escalator in South College" as false.

Does this mean that our Fregean semantics for the definite article has proved empirically inadequate and must be abandoned? Many philosophers and linguists have drawn this conclusion.¹⁵ It is a reasonable conclusion, but it is not inescapable. An alternative response is to reconsider the straightforward identification which we have assumed so far between the semantic values 1 and 0 and the pre-theoretical notions of truth and falsity. Might we perhaps reconcile the present semantic analysis with the empirical evidence if we posit a somewhat more indirect correspondence between the truth-values of our semantic theory and the intuitions that people report in truth-value judgment tasks?

As a first step in this direction, let us propose that the colloquial term "false" covers both truth-value-less sentences and those that are false in the technical sense of denoting 0. In other words, the technical terms of our theory translate into pre-theoretical terms as follows:

$$\begin{array}{ll} \llbracket \phi \rrbracket = 1 & \phi \text{ is true} \\ \left. \begin{array}{l} \phi \text{ has no semantic value} \\ \llbracket \phi \rrbracket = 0 \end{array} \right\} & \phi \text{ is false} \end{array}$$

This stipulation makes the predictions of our semantics consistent with the data we reported above: for example, with the fact that informants who are told that South College contains no escalator and are asked to decide whether (8) is true or false will choose "false".

Mere compatibility with the data is not all we are aiming for, of course. To justify our choice over competing theories, in particular those that make only a 2-way distinction between true and false, we have to show that the additional distinction between two sources of intuitive falsity does some useful work. For instance, we might argue that it helps us to explain certain other manifestations of semantic competence, which can be observed when we move beyond simple truth-value judgment tasks and elicit subtler intuitions.

Indeed, this kind of argument has been offered.¹⁶ Specifically, it has been argued that the technical distinction between lacking a value and denoting 0 can be systematically related to an intuitive distinction: namely, the distinction between what is *asserted* and what is *presupposed*. For an illustration, consider the following three sentences.

- (10) (a) John is absent again today.
 (b) Today is not the first time that John is absent.
 (c) John is absent today, and that has happened before.

All three of these sentences somehow express the speaker's belief that John is absent today and has been absent at least once before. But they are not simply interchangeable. If you are talking to somebody with whom you already share the information that John has been absent in the past, but who doesn't yet know about today, (10a) is a natural choice, but (10b) is not. If you are talking to somebody who already knows that John is absent today, but knows nothing about his past history, then (10b) is natural, whereas (10a) is not. And if your audience knows nothing at all about John's past or present attendance, the most natural choice is (10c). We accordingly say that (10a) *presupposes* that John was absent before and *asserts* that he is absent today. With (10b), it's the other way round: this sentence presupposes that John is absent today and asserts that he was absent before. Finally, (10c) asserts that John both is absent today and was absent before, without presupposing anything.

The ability to discriminate between (10a), (10b), and (10c) and decide which is most appropriate for an audience with a given state of information is clearly part of understanding English. We would therefore like to capture it somehow in our semantics. The hypothesis presently under consideration is that a semantic theory equipped to distinguish two kinds of non-true sentences is better suited to accomplish this than one that isn't. The concrete proposal is that ϕ having no value represents the case that ϕ has a false presupposition, and $\llbracket \phi \rrbracket = 0$ means that ϕ does not presuppose anything false but makes a false assertion. (The third case, $\llbracket \phi \rrbracket = 1$, thus has to mean that both what ϕ presupposes and what it asserts are true.)

Let's return to the definite article. The analysis we presented above (following Frege) may be called a "presuppositional" analysis of "the". In light of what we have said in the present subsection, it predicts that a sentence like (8) ("John is on the escalator in South College") would be used most naturally by a speaker who assumes that her audience knows that there is a unique escalator in South College, but doesn't know about John's whereabouts. This seems basically right. Minimal pairs like (11a, b) below point in the same direction.

- (11) (a) **There will be one mid-term, which will be on November 21st.**
 (b) **The mid-term will be on November 21st.**

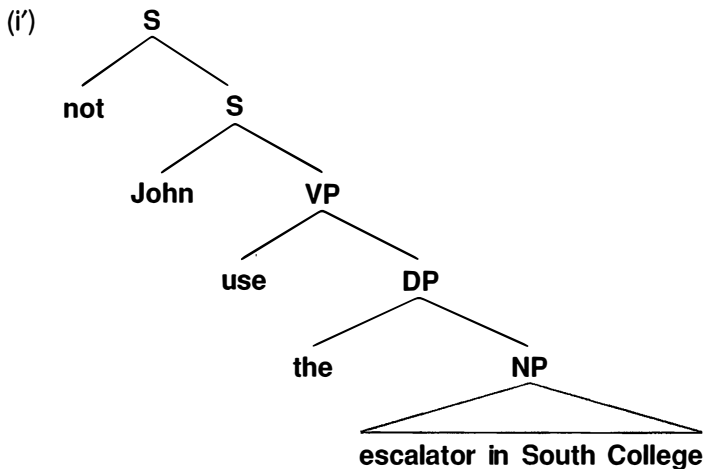
If the topic of mid-term exams for this course hasn't come up yet at all, (11a) is the natural choice; (11b) is fully appropriate only when the audience is already aware that there will be one mid-term. Neither sentence can be true unless there will be a unique mid-term, but in (11a) this is part of the assertion, while in (11b) it is presupposed. Our semantic analysis predicts this: If there isn't a unique mid-term, then (11b) has no truth-value at all. (In contrast with (11a), which should denote 0 in this case. We say "should", because we have yet to develop an analysis of the words and constructions in (11a) that actually predicts this.)

We have made an attempt here to provide some initial evidence in favor of our presuppositional analysis. This is not meant to be a compelling justification of this analysis, however. There are many well-known objections which we have not even mentioned, much less addressed. For the time being (especially for the next chapter), it is convenient to be able to assume *some* concrete analysis of the definite article, even if it turns out to be only a crude approximation. This will allow us to get our analysis of other parts of English off the ground.

Exercise 1

Assume that sentence (i) has the structure (i') at the level at which it is interpreted.

(i) **John doesn't use the escalator in South College.**



In (i'), the determiner "the" is assumed to be a D (determiner, not to be confused with the domain of individuals D, also referred to as " D_e ") heading a DP (determiner phrase).¹⁷ The structure (i') differs from the surface structure of (i) with respect to the position of the subject "John". We might assume that (i') is the Deep Structure of (i), where the subject appears below the negation before movement to a higher position. Or else we could assume that (i') is a Logical Form representation that results from reconstruction of the raised subject into its original position.

What does our current theory predict about this sentence? Are the predictions empirically correct? Construct suitable examples and scenarios that help

sharpen intuitions and promote discussion. It might also be useful to go through Bertrand Russell's "On Denoting" (see n. 15) and see which of his examples would argue against the analysis of the definite article proposed in this chapter.

Exercise 2

Look at the following scenario:

(← LEFT) ☐ 🍏 ☐ 🍏 🍏 ☐ (RIGHT →)
 b₁ b₂ b₃ b₄ b₅ b₆

and consider the following three definite descriptions:

- (i) the leftmost apple in the row
 - (ii) the leftmost dark apple in the row
 - (iii) the apple that is both leftmost in the row and dark
- (a) In your intuitive judgment, which individual, if any, does each of these definite descriptions refer to in this situation?
 - (b) What predictions concerning the denotations of the definite descriptions (i), (ii), and (iii) would follow from an analysis that treats adjectives like "leftmost" as 1-place predicates?
 - (c) Specify a more adequate denotation for attributive uses of "leftmost".
 - (d) In a compositional fashion, compute the denotation of the definite description "the leftmost dark apple in the row", given the above scenario. For the purpose of this computation, take "apple in the row" as an unanalyzed predicate. That is, you don't have to worry about the PP "in the row".
-

4.4.3 Uniqueness and utterance context

Frege's uniqueness presupposition has often been objected to as an idealization that does not really fit the definite singular article in English. We frequently say things like "the door is locked" or "the cat wants to be fed", yet we don't believe that there is just one door and just one cat in the world, and nobody that hears us speak this way will attribute such beliefs to us either.

There are a number of different responses to this objection, but this is not the occasion to give them serious consideration. Somehow or other, we have to

concede that “the cat” doesn’t denote the unique cat that there is in the whole world, but rather denotes, on each occasion on which it is uttered, the unique cat among those individuals that are under consideration on this utterance occasion. Our best attempt for now at making this explicit is in the following revised lexical entry:

- (5') $\llbracket \text{the} \rrbracket =$
 $\lambda f : f \in D_{\langle e, t \rangle}$ and there is exactly one $x \in C$ such that $f(x) = 1$.
 the unique $y \in C$ such that $f(y) = 1$,
 where C is a contextually salient subset of D .

Once again, our lack of a serious account of context-dependency prevents us from stating this more precisely. Below, we will assume something like (5') in informal discussion, but will abstract away from context-dependency and use (5) in our calculations.

4.4.4 *Presupposition failure versus uninterpretability*

In chapter 3, we talked about cases of a rather different sort in which a linguistic expression fails to have a semantic value. Recall our treatment of so-called Θ -Criterion violations like “Ann laughed Jan”. We observed that this sentence is not in the domain of the $\llbracket \rrbracket$ function as defined by our semantic theory. We called such structures “uninterpretable”, and we proposed that the uninterpretability of “Ann laughed Jan” accounted for the ungrammaticality judgment represented by the asterisk.

In the present chapter, however, we have just suggested that sentences which lack a semantic value are intuitively judged as presupposition failures. So it seems that we have not been consistent. Is it a kind of falsity or a kind of ungrammaticality that we want our theory to capture when it provides no denotation for a given structure? The two are obviously quite different intuitively, and by simply conflating them we would be missing a systematic fact about people’s linguistic intuitions. We might try to draw the intended distinction as follows:

- (12) If α is *uninterpretable*, then it can be proved from the semantics alone that α is outside the domain of $\llbracket \rrbracket$.
- (13) If it is a contingent matter of fact that α is outside the domain of $\llbracket \rrbracket$, then α is a *presupposition failure*.

(12) and (13) correctly distinguish between, say, “Ann laughed Jan” on the one hand and “The escalator in South College is moving” on the other. In the former

case, we need not assume anything about the world to show it lacks a denotation. In the latter case, we need to invoke physical facts to show this, and we can easily imagine counterfactual states of affairs in which that sentence *would* have a truth-value.

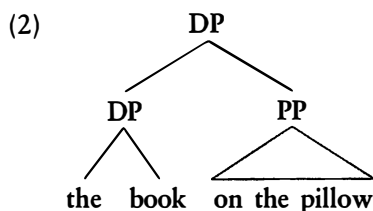
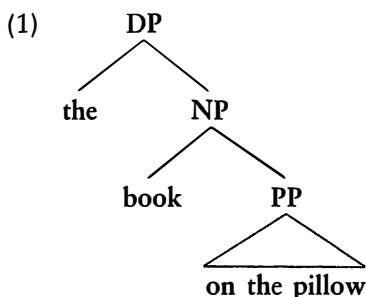
But if we tried to turn (12) and (13) into biconditionals that could stand as *definitions* of uninterpretability and presupposition failure, we would face an objection: namely, that there are sentences which intuitively are “necessary presupposition failures” – for example, “John met the man who died and didn’t die”. By the criteria given in (12), (13), this is indistinguishable from “Ann laughed Jan”: we only need the semantics of English in order to infer that it has no semantic value. But its intuitive status is different, and it should be classified as a presupposition failure rather than as uninterpretable.

So the distinction we are after cannot simply be identified with the difference between necessary and contingent lack of denotation. If we want to characterize it in precise terms, we have to be more specific. In the case of an uninterpretable structure, information about the *type* of each subtree is sufficient to decide that the structure receives no denotation. To detect presupposition failure, by contrast, we must know more about the denotations of certain subtrees than their mere semantic types.

4.5 Modifiers in definite descriptions

We conclude this chapter by highlighting some predictions that our current semantics makes about the interaction of modifiers and the definite article when these co-occur.

As was already implicit in our treatment of examples in section 4.4, we are assuming a syntax according to which restrictive modifiers within DPs (determiner phrases) form a constituent with the head noun to the exclusion of the determiner.¹⁸ That is, the bracketing is as in (1), not as in (2).



Accordingly, our semantics composes the values of the noun and the modifier before composing the result of this with the value of the determiner.

Syntacticians have frequently entertained the opposite hierarchical organization (2). Suppose this were the structure of the object in the sentence "Ann dislikes the book on the pillow". What would happen if we attempted to interpret this sentence by our principles of semantic composition?

The first answer to this question is that the alternative structure in (2) leads to uninterpretability within our framework – not within (2) itself, but in whatever larger structure (2) is part of. What happens is that the lower DP, "the book", denotes the unique (contextually relevant) book (if any), and the higher DP then denotes a truth-value: namely, 1 if the unique book is on the pillow, and 0 if it is not. So any attempt to interpret the next higher node (say, the S-node if (2) is a subject, or the VP-node if it is the object of a transitive verb) will fail due to type-mismatch.¹⁹

In a historic debate, Barbara Partee²⁰ took the uninterpretability of structures like (2) (given certain plausible assumptions) to be an argument against syntactic analyses that imply the bracketing [[determiner noun]modifier] and in favor of those that assume [determiner [noun modifier]]. Noam Chomsky²¹ pointed out in response that there is undeniable evidence for such alleged "uninterpretable" surface bracketings in at least some constructions in some languages. (Directly relevant here, for instance, are Scandinavian languages, where the definite article is realized as a suffix on the noun and intervenes in the linear surface order between it and the PP.) The proper conclusion to draw seems to be that surface structure need not always be the appropriate input to semantic interpretation. Rather, the level that is interpreted is a more abstract one. Our hypothesis then must be that even languages that exhibit the bracketing [[determiner noun]modifier] on the surface have [determiner [noun modifier]] at some other level of representation.

Notes

- 1 For a more systematic treatment of the indefinite article, see ch. 6.
- 2 See especially R. Jackendoff, *X' Syntax* (Cambridge, Mass., MIT Press, 1977).
- 3 Ibid., chs 4 and 7.
- 4 A more explicit formulation along the lines of section 3.3 would be (4').
- (4') If α is a branching node and $\{\beta, \gamma\}$ is the set of α 's daughters, then α is in the domain of $\llbracket \]$ if both β and γ are, and $\llbracket \beta \rrbracket$ and $\llbracket \gamma \rrbracket$ are both in $D_{\langle e, t \rangle}$.
In this case, $\llbracket \alpha \rrbracket = \lambda x \in D_e . \llbracket \beta \rrbracket(x) = \llbracket \gamma \rrbracket(x) = 1$.
- 5 Remember that we couldn't have written " $\llbracket \alpha \rrbracket = \lambda x \in D_e . \llbracket \beta \rrbracket(x)$ and $\llbracket \gamma \rrbracket(x)$ " in the last line of (6). $\llbracket \beta \rrbracket(x)$ and $\llbracket \gamma \rrbracket(x)$ are truth-values. A value description of the

- form "[β](x) and [γ](x)", then, is as ill-formed as a value description of the form "1 and 0".
- 6 Higginbotham calls this operation "theta identification": J. Higginbotham, "On Semantics," *Linguistic Inquiry*, 16 (1985), pp. 547–93, at p. 564.
 - 7 Two influential papers concerning type shifting and flexible types are B. Partee and M. Rooth, "Generalized Conjunction and Type Ambiguity," in R. Bäuerle, C. Schwarze, and A. von Stechow (eds), *Meaning, Use and Interpretation of Language* (Berlin, de Gruyter, 1983), pp. 361–83, and B. H. Partee, "Noun Phrase Interpretation and Type-Shifting Principles," in J. Groenendijk, D. de Jongh, and M. Stokhof (eds), *Studies in Discourse Representation Theory and the Theory of Generalized Quantifiers* (Dordrecht, Foris, 1987), pp. 115–43.
 - 8 For an informative overview and many references, consult C. Hamann, "Adjectival Semantics," in A. von Stechow and D. Wunderlich (eds), *Semantik/Semantics. An International Handbook of Contemporary Research* (Berlin and New York, de Gruyter, 1991), pp. 657–73. Another useful overview article (mainly discussing the semantics of adjectives) is B. H. Partee, "Lexical Semantics and Compositionality," in L. R. Gleitman and M. Liberman (eds), *Language. An Invitation to Cognitive Science*, 2nd edn, vol. 1 (Cambridge, Mass., MIT Press, 1995), pp. 311–60.
 - 9 The best overview article on context dependency (unfortunately in German) is T. E. Zimmermann, "Kontexttheorie," in von Stechow and Wunderlich (eds), *Semantik/Semantics*, pp. 156–229. The references given in Zimmermann's article are a good starting point even for readers who cannot read the article itself. A classic reference is D. Kaplan, "Demonstratives: An Essay on the Semantics, Logic, Metaphysics, and Epistemology of Demonstratives and other Indexicals," in J. Almog, J. Perry, and H. K. Wettstein (eds), *Themes from Kaplan* (New York, Oxford University Press, 1989), pp. 481–563. As for formal accounts of vagueness, we recommend M. Pinkal, *Logic and Lexicon: The Semantics of the Indefinite* (Dordrecht, Kluwer Academic Publishers, 1995). The classic reference for an account of the vagueness of adjectives is H. Kamp, "Two Theories about Adjectives," in E. Keenan (ed.), *Formal Semantics of Natural Language* (Cambridge, Cambridge University Press, 1975), pp. 123–55. See also E. Klein, "A Semantics for Positive and Comparative Adjectives," *Linguistics and Philosophy*, 4 (1981), pp. 1–45.
 - 10 The classic reference is D. Davidson, "The Logical Form of Action Sentences," in N. Rescher (ed.), *The Logic of Decision and Action* (Pittsburgh, University of Pittsburgh Press, 1967), pp. 81–95. See also T. Parsons, *Events in the Semantics of English. A Study in Subatomic Semantics* (Cambridge, Mass., MIT Press, 1990).
 - 11 And we won't even consider the full range of occurrences of this. Since all the common nouns we use in the book are singular count nouns, we say nothing about definite mass and plural terms. We confine our attention to what philosophers call "definite descriptions": viz. singular terms in which "the" combines with an NP headed by a count noun.
 - 12 Frege, "Function and Concept" (1891), trans. in M. Black and P. Geach, *Translations from the Philosophical Writings of Gottlob Frege* (Oxford, Basil Blackwell, 1960), pp. 56–78.
 - 13 Quote from Frege, "Über Sinn und Bedeutung," trans. as "On Sense and Nomination," in A. P. Martinich (ed.), *The Philosophy of Language*, 2nd edn (New York and Oxford, Oxford University Press, 1990), pp. 190–202, at pp. 197–8.
 - 14 If we help ourselves to a little bit of mathematical and logical notation, we can make it shorter:

$\lambda f : f \in D_{\langle e, t \rangle} \ \& \ \exists! x[f(x) = 1] \ . \ \iota y[f(y) = 1]$.

The abbreviatory conventions used here are the following:

" $\exists! x[\phi]$ " abbreviates "there is exactly one x such that ϕ ".

" $\iota x[\phi]$ " abbreviates "the unique x such that ϕ ".

(The first symbol in the last line is the Greek letter iota.)

- 15 Most notably B. Russell, in "On Denoting," *Mind*, 14 (1905), pp. 479–93, who proposed an alternative analysis which predicts that $\llbracket \text{John is on the escalator in South College} \rrbracket = 0$ (given that there is no escalator in South College).
- 16 The classical reference here is once again Frege. See "Über Sinn und Bedeutung," pp. 196–7.
- 17 See S. Abney, "The English Noun Phrase in its Sentential Aspect" (Ph.D. dissertation, MIT, 1987).
- 18 The issue did not really arise before section 4.4, since in section 4.3 we only used the determiner "a". Since that was taken to be vacuous, nothing depended on its exact place in the phrase structure hierarchy.
- 19 We predict, thus, that if structures like (2) are generated by the syntax of English at all, they can only be used unembedded to make statements, or embedded in positions where truth-values are selected (e.g., as complements to truth-functional connectives). This possibility seems not to be realized, a fact for which we must seek some syntactic explanation.
- 20 B. H. Partee, "Some Transformational Extensions of Montague Grammar," in B. Partee (ed.), *Montague Grammar* (New York, Academic Press, 1976), pp. 51–76, esp. pp. 53ff. Actually, Partee makes her point about restrictive relative clauses rather than restrictive PP modifiers. But the internal semantics of relative clauses (see next chapter) is not relevant to it, and it carries over to any kind of postnominal restrictive modifier.
- 21 N. Chomsky, "Questions of Form and Interpretation," *Linguistic Analysis*, 1/1 (1975), pp. 75–109, esp. pp. 96ff.